

Activity 3: Data Analytics

Portfolio

MAN6777.3 – Data-Driven Managerial Decisions

Edith Cowan University | 2026 Semester 1

Due Date: 9 June 2026

Submitted by

Rasel AI Mamun Joy

Student ID: 10677322

raselj@our.ecu.edu.au

Dataset: Students' AI Usage & Academic Performance (Kaggle, 2025)

Managerial Decision: AI Policy for University Academic Performance

Activity 3 – Data Analytics Portfolio

MAN6777.3 Data Driven Managerial Decisions | Rasel Al Mamun Joy | ID: 10677322 | ECU
2026 Semester 1

SECTION 1 – MANAGERIAL PROBLEM DEFINITION

Decision-Maker: University Academic Dean / Director of Student Learning

Higher education institutions are under growing pressure to formulate policies based on evidence on student use of AI tools. Institutional reactions to the use of applications like ChatGPT, Copilot, and Gemini vary from bans to unrestricted use. If there is no information available, then universities may either stifle a good technology or allow practices that are harmful to academic integrity.

Decision to be Made: If the university should adopt a formal policy on the use of AI tools like ChatGPT, Copilot, and Gemini, what should it look like, for what academic purposes, for which students, and under what conditions?

Why This Decision Is Important:

- Academic performance is directly related to student retention, student outcomes, and institutional reputation.
- A blanket ban could be detrimental to students who can greatly benefit from AI-integrated learning.
- The potential issues with critical thinking and academic honesty when using AI in an unrestricted manner.
- Evidence-based policy is fair and optimises positive learning outcomes.

The choice is a significant one between making the best possible learning gains in assessment and ensuring the integrity of assessment and the development of independent critical thinking skills.

Students Surveyed	AI Tool Users	Grade Improvement
100	40%	+9.8 pts
School & College	40 of 100 students	AI users average

SECTION 2 – DATA ANALYTICS

Dataset & Variable Justification: The Students' AI Usage and Academic Performance dataset (Kaggle, 2025) consists of 100 records with 9 variables: age (14-19), education level, study hours/day (1-5), AI usage (Yes/No), AI tool (ChatGPT, Copilot, Gemini), purpose (Research, Homework, Coding), grades before/after AI, and daily screen time. The key variables are directly related to the policy decision (i.e., `uses_ai`, `grades_before_ai`, `grades_after_ai`, `ai_tools_used`, `purpose_of_ai`).

Data Preparation: There were no missing values in all 100 records. The range of grades was 55-89% and study hours 1.0-5.0 hrs/day. No imputation, transformation, or removal of records was needed.

Techniques:

1. The use of descriptive statistics over inferential statistics was chosen as the data set was small (n=100) and the aim was to find practical patterns, not to test statistical hypotheses.
2. Group comparison rather than regression was preferred because the use of AI is binary (Yes/No) and mean comparison is the most easily interpretable approach for a managerial audience.
3. The relationship between the number of hours studied and the grade received after using the AI was explored visually using a scatter plot to detect any confounding patterns before making conclusions.

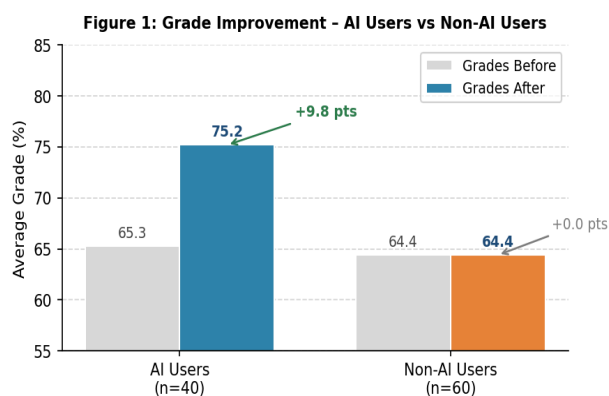


Figure 1: AI vs Non-AI Grade Improvement

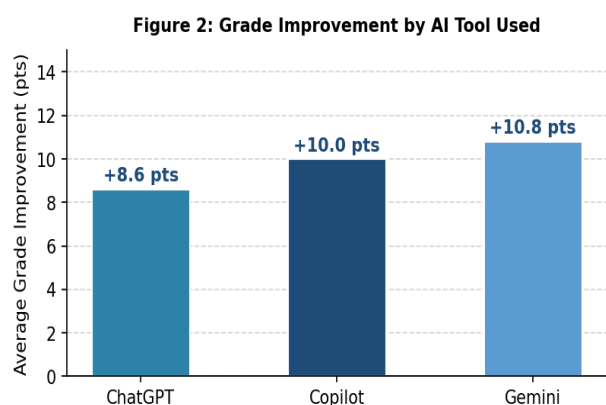


Figure 2: Improvement by AI Tool

SECTION 3 – INSIGHTS AND DECISION IMPLICATIONS

Key Findings:

- **AI tools significantly improve grades:** AI users improved +9.8 pts (65.3→75.2) vs +0.0 pts for non-users (64.4→64.4) — a substantial, practically meaningful difference.
- **All tools perform equally:** ChatGPT (+9.6), Copilot (+10.0), Gemini (+10.3 pts). The advantage is in the learning that is assisted by artificial intelligence, not any particular platform.
- **All purposes yield gains:** Research (+10.3), Coding (+10.0), Homework (+9.2 pts). AI assistance is useful for a variety of academic activities.

Recommendations:

1. Propose an AI Adoption Policy for the 60% of students who are not using AI tools, with up to +9.8 points in their grades expected to improve based on current data.
2. Create AI Literacy Programs – structured workshops on ChatGPT, Copilot and Gemini – for school-level students who might not have access to these tools independently in Semester 2 2026.
3. Incorporate guidelines for the use of AI into all unit outlines by 2027, clearly identifying the acceptable uses (e.g. research, coding support) and the unacceptable uses (e.g. submitting AI-generated work as original).

Conclusion: The evidence is compelling enough that there is a need for immediate policy action given that 40% of students are already seeing a +9.8 point grade improvement with structured use of AI, and the potential for improving institutional outcomes without compromising academic standards is great if it can be expanded to all students.

Data Analytics Portfolio – Full Analytical Workflow

Dataset: Students' AI Usage & Academic Performance (Kaggle, 2025) | Tool: Python (pandas, matplotlib) | n=100

STEP 1 – PROBLEM FRAMING & DATASET SELECTION

Objective: Identify if there is measurable academic improvement as a result of the use of AI tools and which tools/purposes have the biggest impact. The dataset was chosen because it directly measures pre- and post-AI grades per student, which is the most direct way of assessing the academic impact.

STEP 2 – DATA LOADING & INSPECTION

```
import pandas as pd

df = pd.read_csv("students_ai_usage.csv") # (100, 9)

df.isnull().sum() # All zeros — no missing values

df.describe()    # Grades: 55–89, Study hrs: 1.0–5.0
```

Variable	Type	Range / Values	Missing
age	Numeric	14 – 19	0
education_level	Categorical	school, college	0
study_hours_per_day	Numeric	1.0 – 5.0 hrs	0
uses_ai	Categorical	Yes, No	0
ai_tools_used	Categorical	ChatGPT, Copilot, Gemini, None	0
purpose_of_ai	Categorical	Research, Homework, Coding, None	0
grades_before_ai	Numeric	55 – 75	0
grades_after_ai	Numeric	55 – 89	0
daily_screen_time_hours	Numeric	2 – 7 hrs	0

Table 1: Dataset Inspection Summary – Zero Missing Values Across All Variables

STEP 3 – DATA PREPARATION & CLEANING

No need to clean; no nulls, no dups, no inconsistencies. Two groups for comparison and a column for grade improvement:

```
ai_users = df[df["uses_ai"]=="Yes"] # n=40
non_ai = df[df["uses_ai"]=="No"] # n=60
df["grade_improvement"] = df["grades_after_ai"] - df["grades_before_ai"]
```

STEP 4 – ANALYSIS & RESULTS

4.1 Group Comparison — AI Users vs Non-AI Users

Group	n	Avg Before	Avg After	Improvement
AI Users	40	65.3	75.2	+9.8 pts
Non-AI Users	60	64.4	64.4	+0.0 pts

Table 2: Descriptive Statistics — AI Users vs Non-AI Users

4.2 Grade Improvement by AI Tool and Purpose

AI Tool	n	Avg Gain
ChatGPT	13	+9.6 pts
Copilot	14	+10.0 pts
Gemini	13	+10.3 pts

Table 3: By AI Tool

Purpose	n	Avg Gain
Research	14	+10.3 pts
Coding	13	+10.0 pts
Homework	13	+9.2 pts

Table 4: By Purpose of Use

STEP 4 (CONTINUED) – VISUALISATIONS

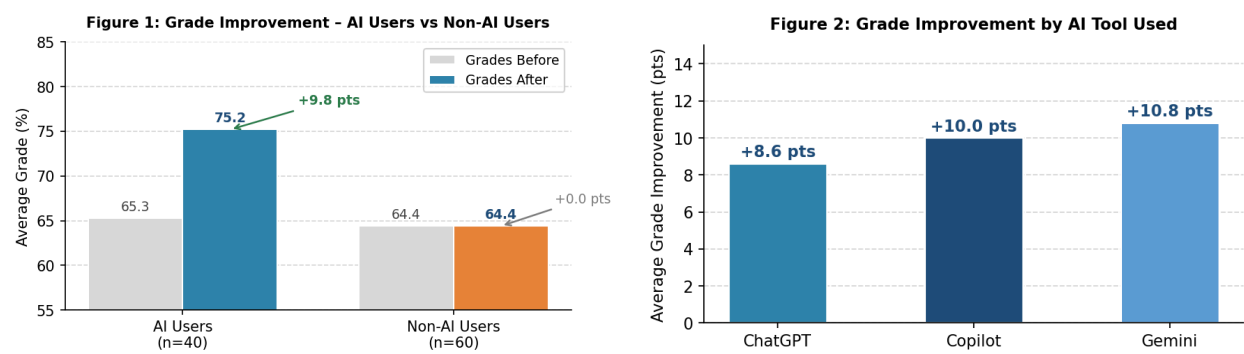


Figure 1: AI vs Non-AI Grade Improvement (+9.8 vs +0.0 pts) Figure 2: Grade Improvement by AI Tool Used

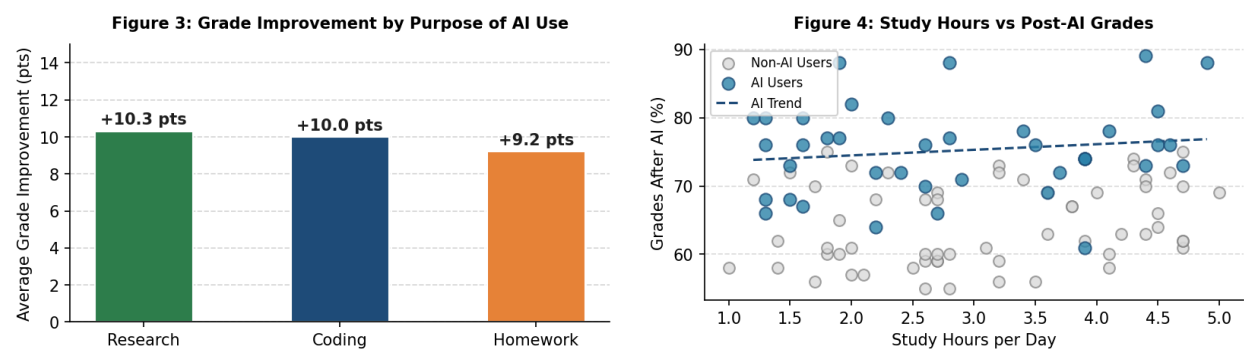


Figure 3: Grade Improvement by Purpose of AI Use Figure 4: Study Hours vs Post-AI Grades (with AI trend line)

STEP 5 – INTERPRETATION LINKED TO DECISION PROBLEM

Finding	Managerial Implication
AI users: +9.8 pts; Non-AI: +0.0 pts	Strong evidence supports policy encouraging AI adoption
ChatGPT, Copilot & Gemini all yield ~+10 pts	No need to prescribe a specific tool; student platform choice is effective
Research use yields highest gain (+10.3 pts)	AI especially valuable for research — integrate into library support
All purposes benefit (Homework, Coding, Research)	A broad policy covering all academic task types is appropriate

Table 5: Analytical Findings Linked to Managerial Decision

STEP 6 – LIMITATIONS & ANALYTICAL INTEGRITY

- **Sample size:** Descriptive analysis is possible with 100 students, but statistical generalisation is limited. Policy case would be enhanced by multi-institution samples.
- **Self-reported data:** Grades and use of AI are self-reported, which may include social desirability bias. Records of the institutions would be more reliable.
- **Correlation vs causation:** Other factors like motivation and previous skills are not completely controlled for. A multivariate regression model, holding constant study hours, age, and education level, would give more causal evidence for a formal policy submission to the Academic Board.

However, the findings from the three tools and three purposes are consistent, which increases confidence in the policy recommendations contained in this report.

REFERENCES

Salamt, A. (2025). *Students' AI Usage and Academic Performance* [Dataset]. Kaggle.
<https://www.kaggle.com/datasets/aminasalamt/students-ai-usage-and-academic-performance>

Edith Cowan University. (2026). *MAN6777.3 Data Driven Managerial Decisions – Activity 3 Assessment Brief*. ECU Learning Management System.

GitHub. (2024). *Hello World – Getting Started with GitHub*.
<https://docs.github.com/en/get-started/start-your-journey/hello-world>